

4 (a) Out of paper.

(b) (i) The magnetic force acting on the α -particle provides the centripetal force for its circular motion.

$$Bqv = \frac{mv^2}{r}$$

$$B(2e)v = \frac{mv^2}{r}$$

$$2Be = \frac{mv}{r}$$

$$2Be = p/r$$

$$p = 2Ber$$

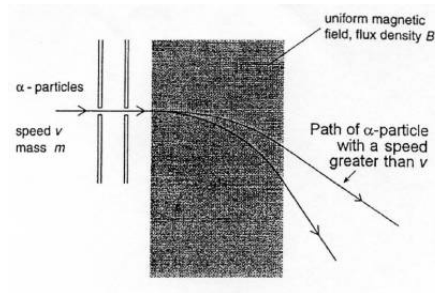
$$(ii) KE = \frac{1}{2}mv^2$$

$$= \frac{(mv)^2}{2m}$$

$$\begin{aligned}
 &= \frac{p^2}{2m} \\
 &= \frac{(2Ber)^2}{2m} \\
 &= \frac{2(Be)^2}{m}
 \end{aligned}$$

(c) Rearranging the equation in (b)(i), the radius of curvature $r = \frac{p}{2Be}$

An α -particle with a greater speed will have a greater momentum p and hence a greater radius of curvature r within the magnetic field. Once outside the magnetic field, the resultant force acting on the α -particle is zero. Newton's first law of motion applies and the α -particle travels in a straight



Note: Some errors include having the line continued as a curve when outside the area of the field.